α1-Antitrypsin Is Degraded and Non-Functional in Chronic Wounds But Intact and Functional in Acute Wounds: The Inhibitor Protects Fibronectin from Degradation by Chronic Wound Fluid Enzymes

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Fluid obtained from chronic and acute wounds were examined for the presence of fibronectin, α1-antitrypsin, and proteinases capable of degrading both proteins. Immunoblot analysis of fluids from ten chronic wounds revealed that fibronectin and α1-antitrypsin were degraded in nine of ten samples. In contrast, both fibronectin and α1-antitrypsin were intact in acute wound fluids. The degradation of the inhibitor and fibronectin occurred in the same wound fluids, and these two events correlated perfectly. Chronic or acute wound fluid proteins were coupled to benzamidine Sepharose 6B beads and incubated with fibronectin or α1-antitrypsin. Chronic wound fluid proteins degraded fibronectin in the presence of ethylenediaminetetraacetate, leupeptin, cystatin, and pepstatin but not in the presence of phenylmethylsulfonyl fluoride. Acute wound fluids and normal human serum did not contain enzymes capable of degrading fibronectin. These data suggest that serine proteinases are responsible for fibronectin degradation in chronic wound fluids. Chronic wound fluids that contained degraded α1-antitrypsin also contain proteinases capable of degrading α1-antitrypsin from human serum. Acute wound fluids and normal human serum did not contain enzymes capable of degrading α1-antitrypsin. The inhibitor from acute wound fluids bound to one of its targets, trypsin. In contrast, the fragment(s) of α1-antitrypsin from chronic wound fluids did not bind trypsin. Chronic wounds associated with degraded fibronectin and the inhibitor contained ten- to forty-fold more elastase activity than acute wounds. The degradation of fibronectin by chronic wound fluid enzymes was inhibited by α1-antitrypsin in a dose-dependent manner. Collectively, these results demonstrate that there are enzymes in chronic wounds that perturb the function of α1-antitrypsin and allow fibronectin degradation by uninhibited serine proteinases. ***Key words: serine proteinases. J Invest Dermatol 105:572–578, 1995***

Fibronectin (FN) and fibrin constitute the major components of the provisional matrix in skin wounds [1–4]. These components participate in the repair process and provide substrata for migrating keratinocytes, fibroblasts, and endothelial cells. FN is proposed to be a beneficial factor in the healing of skin wounds. In some chronic, non-healing wounds FN is extensively degraded [5,6]. The enzymes responsible for FN degradation in chronic wounds remains unclear. It has been reported that chronic wound fluids contain increased amounts of matrix metalloproteinases (MMPs)-2 and -9 and are involved in the degradation of FN in chronic wounds [7]. But in burn-wound fluids, Grinnell and Zhu reported that neutrophil elastase, not MMPs, are responsible for FN degradation [8].

The aim of the experiments reported herein was to investigate if serine proteinases are responsible for FN degradation in chronic skin wounds and to examine the status of α1-antitrypsin (AT) (also called α1-protease inhibitor) [9], in acute and chronic wounds. The results suggest that serine proteinases, not MMPs, are responsible for FN degradation in chronic wounds. Elastase activity is ten- to forty-fold higher in fluids from chronic wounds compared to acute wounds and normal human serum (NHS). The results also implicate AT as a regulator of FN degradation in chronic wounds. The inhibitor protects FN from degradation by serine proteinases in chronic wounds. AT is intact and functional in acute wounds but is degraded and inactive in chronic wounds. AT-degrading enzymes were present in chronic wounds associated with degraded AT but absent in wound fluids with intact AT. Based on these observations, we propose that the loss of functional AT is responsible for FN degradation in chronic wounds.

MATERIALS AND METHODS

Collection of Chronic and Acute Wound Fluids Patients undergoing treatment for chronic venous stasis ulcers (greater than 6 months duration) were the source of chronic wound fluids used in this study. The procedure

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Abbreviations: AT, α1-antitrypsin; BZ-6B, benzamidine sepharose 6B; FN, fibronectin; MMPs, matrix metalloproteinases; NHS, normal human serum.

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for the collection of chronic wound fluids was as follows. The wound dressing containing the wound fluid was removed from the patient during the dressing change. The fluid from the wound dressing was expressed mechanically. If the fluid could not be harvested that day, the dressing was stored at −80°C, thawed to room temperature, and then harvested. If the fluid from the dressing was minimal, 1–10 ml of normal saline was added to the dressing and after 2 min the added fluid was mechanically expressed. Acute wound fluid was obtained from aspirates of normal skin graft donor sites covered with occlusive dressings, or from suction drains after breast surgery.

The wound fluids were diluted tenfold with 15 mM Tris-HCl, pH 7.4, and centrifuged in a microcentrifuge for 10 min (12,000 × *g* at 4°C). The clear supernatant was removed and an aliquot of the sample was analyzed for protein determination by the method of Bradford [10]. The typical sample ranged from 15 to 40 mg protein per milliliter of wound fluid.

Sodium Dodecylsulfate–Polyacrylamide Gel Electrophoresis (SDS-PAGE) and Immunoblotting Proteins were separated by SDS-PAGE using 12% mini-gels under non-reducing conditions [11].

Proteins were electroblotted onto nitrocellulose membranes as described before [12]. After blocking in 4% non-fat dry milk in 10 mM Tris-HCl, 150 mM NaCl, pH 7.4, containing 0.1% Tween-20 (TTBS) for 2 h at room temperature, the membranes were incubated for 2 h in pre-immune serum or rabbit polyclonal antibodies to human AT (affinity-purified IgG, Sigma Chemical Co., St. Louis, MO) or goat polyclonal antibodies to human plasma FN (purified IgG, Cappel Laboratories, Organon-Teknika Corp., West Chester, PA) diluted 1:500 in TTBS. After three 5-min washes, the membranes were incubated for 1 h with peroxidase-conjugated secondary antibody (affinity-purified IgG–horseradish-peroxidase–conjugated, goat anti-rabbit for FN and rabbit anti-goat for AT, Bio-Rad Research Laboratories, Richmond, CA) diluted 1:1000 in TTBS. The antibody-binding proteins were identified by using the 4-chloro-napthal membrane peroxidase substrate system (Kirkegaard and Perry Laboratories, Gaithersburg, MD).

Coupling of Wound Fluid Proteins to Benzamidine Sepharose 6B Two hundred fifty microliters of packed benzamidine Sepharose 6B (BZ-6B) beads (Pharmacia-LKB Biotechnology Inc., Uppsala, Sweden; lots 214844 and 219956) were washed four times with 15 mM Tris-HCl, pH 7.40 (incubation buffer). The beads were incubated with 150 μl wound fluid sample or NHS in 1.5 ml of incubation buffer for 1 h at room temperature. The beads were pelleted by brief centrifugation in a microcentrifuge, washed three times with incubation buffer, and resuspended in 15 mM Tris-HCl, pH 7.40, 5 mM $CaCl_2$, 5 mM $MgCl_2$.

Degradation of FN by Wound Fluid Proteins in the Presence or Absence of Proteinase Inhibitors BZ-6B complexed with proteins from wound fluids and NHS were resuspended in 400 μl of 15 mM Tris-HCl, pH 7.4, 5 mM $CaCl_2$, 5 mM $MgCl_2$, 15 mM ethylenediaminetetraacetic acid (EDTA) or 15 mM Tris-HCl, pH 7.40, 5 mM $CaCl_2$, and 5 mM $MgCl_2$ containing one of the following inhibitors: 20 μg cystatin, 20 μg leupeptin, 20 μg pepstatin, or 10 mM phenylmethylsulfonyl fluoride (PMSF). When these four inhibitors were used, they were added 20 min prior to the addition of 10 to 30 μg of human plasma FN (New York Blood Center, NY). The reaction mixtures were incubated for different periods of time or for 6 h as described in figure legends. At the end of each time point, the supernatant was collected by centrifugation of the beads, diluted to 1 × with 5 × SDS-PAGE sample buffer and FN fragments were detected by immunoblotting with anti-FN antibodies or SDS-PAGE with silver stain. At the end of some experiments, the beads were resuspended in 2 × SDS-PAGE sample buffer and the extract analyzed by immunoblotting with anti-FN antibodies, as described above.

Trypsin Affinity Chromatography of Chronic and Acute Wound Fluid Proteins Trypsin (2 mg/ml) was coupled to Reactigel agarose beads following the manufacturer's instructions (Pierce Chemical Co., Rockford, IL). Five hundred microliters of 1:10 diluted NHS or wound fluid samples were incubated with 100 μl of trypsin-Reactigel beads that were previously equilibrated in 15 mM Tris-HCl, pH 7.4 (equilibration buffer). The incubation was continued for 1 h at room temperature with end-over-end rotation. The supernatant was removed and the trypsin-Reactigel beads were washed three times with equilibration buffer. Trypsin bound proteins were extracted into 150 μl of Laemmli's 2 × SDS-PAGE sample buffer without β-mercaptoethanol. AT or fragments of AT were detected by immunoblotting with anti-AT antibody.

Degradation of FN by Proteins from Chronic Wound Fluids in the Presence of AT BZ-6B (50 μl packed volume) complexed with chronic wound fluid proteins were resuspended in 200 μl of 15 mM Tris-HCl, pH 7.4, 5 mM $CaCl_2$, 5 mM $MgCl_2$ containing 25 to 125 μg AT (Sigma). After a 20-min incubation with the inhibitor, 15 μg of FN was added to the reaction mixture and incubated for 8 h with end-over-end rotation. Supernatants were collected by centrifugation, diluted to 1 × with 5 × SDS-PAGE sample buffer, and FN polypeptides were detected by immunoblotting with anti-FN antibody, as described above.

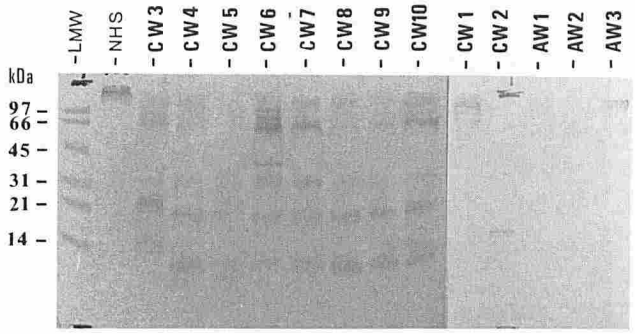


Figure 1. Extensive degradation of FN in chronic wound fluids (CW) but not acute wound fluids (AW). Twenty-five micrograms of proteins from NHS, acute, and chronic wound fluids were separated on 12% polyacrylamide gels, transferred to nitrocellulose membranes, and immunoblotted with goat polyclonal antibodies to FN. The details of immunoblotting are described in *Materials and Methods*.

Degradation of AT by Enzymes from Chronic Wound Fluids One hundred microliters of BZ-6B complexed with proteins from NHS, acute, or chronic wound fluids were incubated in 350 μl of Tris-HCl, pH 7.40, 5 mM $CaCl_2$, 5 mM $MgCl_2$ containing 400 μg of normal human serum proteins (as the source of AT). The beads were allowed to react with AT for different periods of time. Seventy-microliter portions of the clear supernatant were collected at 0 min, 30 min, 2 h, 4 h, and 20 h. The supernatant was adjusted to 1 × by diluting the sample with 5 × SDS-PAGE sample buffer without β-mercaptoethanol. Fragments of AT were detected in 30 μl portions of the supernatants by immunoblotting with anti-AT antibody.

Quantitation of Elastase in Wound Fluids Elastase levels were quantitated in NHS, acute, and chronic wound fluids as described before [8] using methoxysuccinyl-ala-ala-pro-val-p-nitroanilide (Sigma) as the substrate. One hundred micrograms of proteins from wound fluid samples or 0.05 mU to 0.5 mU of human leukocyte elastase (EC 3.4.21.37., Sigma) were incubated in 0.1 M HEPES buffer, pH 7.30, containing 0.5 M NaCl, 10% dimethylsulfoxide, and 0.1 mM elastase substrate for 1 h at room temperature. Substrate degradation was measured at 410 nm using a Beckman 640 spectrophotometer. Wound fluid elastase levels were read against a standard curve obtained with leukocyte elastase and are expressed as milliunits per milligram of wound fluid proteins.

RESULTS

Extensive Degradation of FN in Chronic Wounds Proteins from three acute wounds, ten chronic wounds, and NHS were subjected to SDS-PAGE and immunoblotting against anti-FN antibodies. Under non-reducing conditions, purified plasma FN has a molecular size of 440 kDa and barely penetrated the 12% polyacrylamide gels (data not shown). Like NHS, acute wound fluids largely contained intact FN (**Fig 1**, lanes marked NHS, AW1, AW2, AW3 and AW4). In contrast to the acute wound fluids, nine of ten samples of proteins from chronic wounds contained FN fragments with molecular sizes between 20 and 140 kDa (**Fig 1**, CW2-CW10). The only exception was chronic wound fluid marked CW1, which contained largely intact FN (lane CW1). FN was also intact in three blister fluids and one additional acute wound fluid (data not shown). These results confirm those of Grinnell and co-workers [5,6], who also reported that FN was largely intact in acute wounds whereas it was extensively degraded in some chronic wounds.

Degradation of FN by Chronic Wound Fluid Enzymes Was Not Inhibited by EDTA The enzymes in chronic wounds thought to be responsible for FN degradation are MMPs 2 and 9 [7]. However, we found no correlation between FN degradation

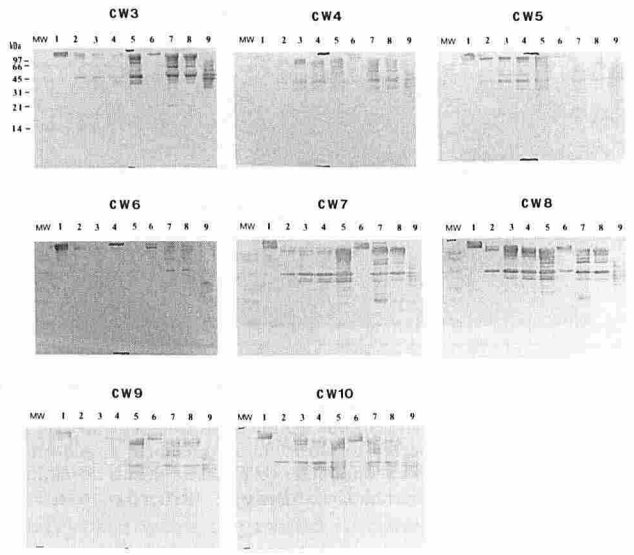


Figure 2. Chronic wound fluids (CW) contain serine proteinases that are responsible for the degradation of human plasma FN. Proteins from CW3, CW4, CW5, CW6, CW7, CW8, CW9, and CW10 were coupled to benzamidine Sepharose 6B (BZ-6B). The beads were incubated with human plasma FN (25 μg) in a total volume of 400 μl containing 15 mM Tris-HCl, pH 7.4, 5 mM $CaCl_2$, 5 mM $MgCl_2$, 10 mM EDTA (for lanes 1–5 in all the panels) at room temperature. Eighty-microliter aliquots of the supernatants were removed at various time points to which 20 μl of 5 × SDS-PAGE sample buffer was added. A 30-μl aliquot was analyzed by immunoblotting with polyclonal antiserum to FN. In all the panels, *lanes 1 to 5* represent the time of incubation: *lane 1*, 0 min; *lane 2*, 30 min; *lane 3*, 120 min; *lane 4*, 240 min; and *lane 5*, 16 h. The ability of PMSF, cystatin, leupeptin, and pepstatin to inhibit the degradation of FN by BZ-6B containing chronic wound fluid enzymes was performed in incubation buffer without EDTA. The degradation of FN was terminated after 6 h. *Lanes 6 to 9* represent the following inhibitors: *lane 6*, 10 mM PMSF; *lane 7*, 20 μg cystatin; *lane 8*, 20 μg leupeptin; and *lane 9*, 20 μg pepstatin.

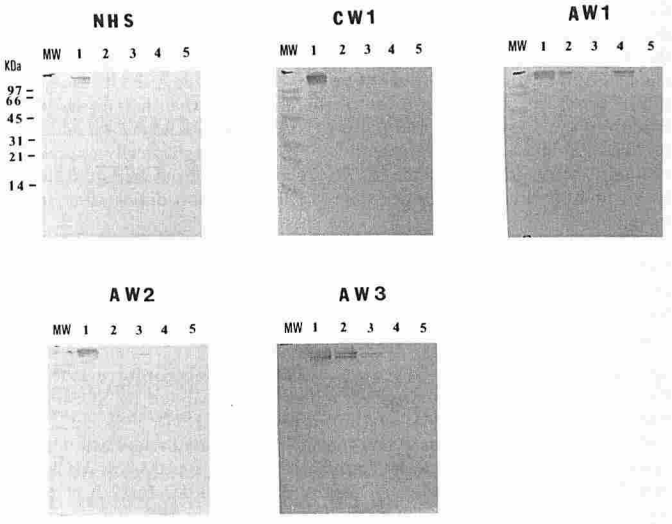


Figure 3. Acute wound fluids AW1, AW2, and AW3, normal human serum, and CW1 do not contain enzymes capable of degrading FN. The experimental conditions were identical to those in **Fig 2**. *Lane 1*, 0 min; *lane 2*, 30 min; *lane 3*, 120 min; *lane 4*, 4 h; and *lane 5*, 16 h after incubation of FN.

and the gelatinolytic activities of pro- and activated MMP-2 and -9 in acute and chronic wound fluids (Rao, unpublished observations). To demonstrate that MMPs are not responsible for FN degradation in chronic wounds, the degradation of FN by chronic wound fluid enzymes was performed in the presence of EDTA, a potent inhibitor of MMPs. Proteins from acute and chronic wounds and NHS were coupled to BZ-6B beads and incubated with FN in presence of EDTA. Degradation of FN occurred by BZ-6B beads complexed with proteins from nine of ten chronic wound fluids. At 0 min, FN was intact (**Fig 2**, lane 1 of all panels). Incubation of FN with these beads for 30 min (lane 2), 2 h (lane 3), 4 h (lane 4), and 16 h (lane 5) resulted in the appearance of specific FN fragments (20–120 kDa) that progressively increased (lanes 2 through 5 of all panels). Identical results were obtained when the degradation of FN was carried out in the absence of EDTA (data not shown). FN fragments were also detected by SDS-PAGE with a sensitive silver stain reagent. By this technique, FN was fragmented into polypeptides of 14, 26, 28, 31, 34, 39, 40, 45, and 55 kDa (data not shown). Collectively, these results demonstrate that MMPs are not responsible for FN degradation in chronic wound fluids associated with degraded FN.

FN Degrading Proteinases Are Not Present in Acute Wound Fluids, CW1, and NHS When proteins from acute wound fluids, NHS, and CW1 were coupled to BZ-6B, they failed to degrade FN. No FN fragments were detected (**Fig 3**, lanes 2–5). In some samples FN is lost with increasing time of incubation (lanes 3 to 5). In all these cases, intact FN was detected on BZ-6B beads (data not shown). Similar results were obtained when these experiments were carried out in the absence of EDTA (data not shown). The possibility of any FN fragments that cannot be detected by anti-FN antibody was also considered. No FN fragments were detected by SDS-PAGE with a sensitive silver stain reagent (data not shown). Thus, acute wound fluids, NHS, and CW1 did not contain FN degrading enzymes.

Serine Proteinases in Chronic Wound Fluids Are Responsible for FN Degradation To identify enzymes responsible for FN degradation in chronic wound fluids, degradation of FN by chronic wound fluid proteins was performed in the presence of serine-, cysteine-, and acid-proteinase inhibitors. Incubation of FN with proteins from chronic wound fluids in the presence (**Fig 2**, all panels, lanes 6–9) of these inhibitors was allowed for 6 h. FN degradation was completely inhibited by 10 mM PMSF (**Fig 2**, lane 6 of all panels) but no inhibition occurred in the presence of 20 μg of cystatin, leupeptin, and pepstatin (**Fig 2**, lanes 7, 8, and 9 of all panels, respectively). Because PMSF is a specific inhibitor of serine proteinases [13], we conclude that this class of enzymes is responsible for FN degradation in chronic wound fluids. Further, leupeptin, which is an inhibitor of trypsin-like proteinases, failed to inhibit the degradation of FN by chronic wound fluid enzymes. This observation points to an elastase- or chymotrypsin-like enzyme likely responsible for FN degradation in chronic wounds. These observations are in agreement with those of Grinnell and Zhu [8] who reported that a serine proteinase, preferably an elastase- or chymotrypsin-like enzyme, is responsible for FN degradation in burn-wound fluids.

AT Is Degraded in Chronic Wound Fluids But Intact in Acute Wound Fluids Because a serine proteinase class of enzymes is responsible for FN degradation in chronic wound fluids, we sought to investigate if serum serine proteinase inhibitors (serpins), particularly AT [9], were intact and functional in chronic wounds.

As shown in **Fig 4**, AT was degraded to a major 37-kDa polypeptide in all chronic wounds in which degraded FN was observed whereas the AT was intact in wounds without degraded FN. Like AT, other serpins, namely, antichymotrypsin, antithrombin III, and α2-antiplasmin, were also degraded in wound fluids in which there was degraded FN and intact in which FN was largely intact (Rao, unpublished data). In a number of lanes (**Fig 4**) particularly with chronic wound fluid samples, anti-AT antibodies also labeled proteins of 90 kDa and higher. Most likely, these are

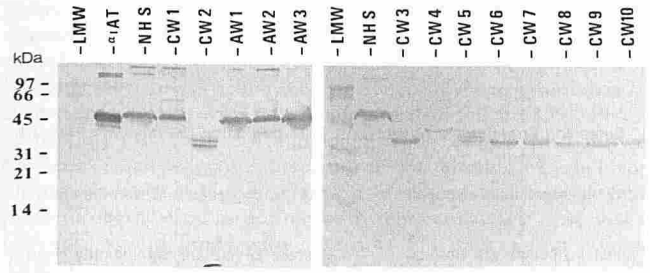


Figure 4. α1-antitrypsin (AT) is degraded in chronic wound fluids (CW) but intact in acute wound fluids (AW). Twenty micrograms of proteins from NHS, acute, and chronic wound fluids were transferred from 12% polyacrylamide gels to nitrocellulose membranes and immunoblotted with rabbit polyclonal antibodies to human AT.

aggregates of AT, as it was known to polymerize, which was also observed with pure AT (lane marked AT).

Trypsin Does Not Bind Fragments of AT in Chronic Wound Fluids Although AT was not intact in chronic wounds associated with degraded FN, AT fragments were observed. To determine if these fragments are capable of binding to trypsin, we incubated 1.5 mg of acute and chronic wound fluid proteins with trypsin-Sepharose. Trypsin is one of the targets of AT [9]. Any bound inhibitor was identified by immunoblotting the trypsin-Sepharose bound proteins against anti-AT antibodies. Acute wound fluids NHS and CW1 subjected to immunoblotting produced a 50-kDa band labeled by the anti-AT antibodies **(Fig 5)**. In contrast, fragment(s) of AT from CW4, CW5, and CW2 did not bind trypsin **(Fig 5)**. In CW3, CW6, CW7, CW8, and CW10, the amount of the 37-kDa AT fragment and AT recovered from trypsin-Sepharose was seven- to tenfold less compared to the AT from NHS and acute wound fluids. Therefore, these experiments suggest that AT is functional in acute wounds whereas the fragments of AT in chronic wounds associated with degraded FN were non-functional.

Degradation of FN by Chronic Wound Fluid Proteinases Was Inhibited by AT To assess which serpin protects FN from degradation by chronic wound fluid enzymes, AT, antichymotrypsin, antithrombin III, and α2-antiplasmin were added to reaction mixtures containing FN and BZ-6B bound proteins from chronic wound fluids. Incubation of FN for 8 h with the BZ-6B bound proteins from six chronic wounds resulted in the appearance of FN fragments ranging between 120 kDa and 10 kDa (**Fig 6**, lane 2 of all panels). In contrast, when the degradation of FN was carried out in the presence of 5 μg (all panels, lane 3) and 20 μg (all panels, lane 4) of AT, it was protected from degradation. Evidence for the inhibition of FN degradation by AT were twofold: 1) intact FN was

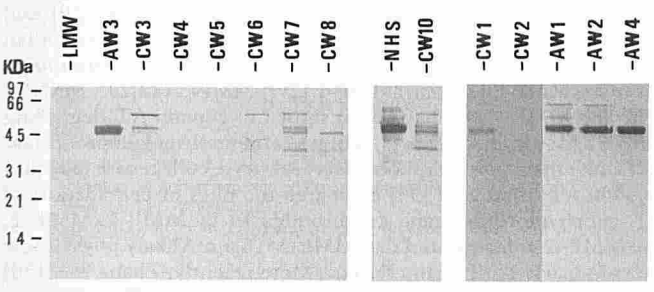


Figure 5. Thirty-seven kilodalton fragment of AT from chronic wound fluids (CW) does not bind trypsin. CW and acute wound (AW) fluids were allowed to bind trypsin and the bound proteins were eluted into 150 μl of SDS-PAGE sample buffer. An aliquot (30 μl) was analyzed by 12% SDS-PAGE and immunoblotted with polyclonal antibodies to AT. The details are in *Materials and Methods*.

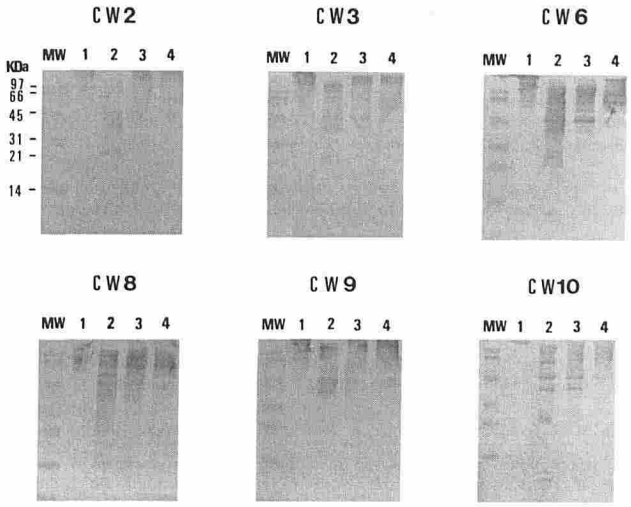


Figure 6. AT inhibits the degradation of plasma FN by chronic wound fluid enzymes. BZ-6B containing proteins from CW2, CW3, CW6, CW9, and CW10 were incubated with increasing amounts (30 μg and 120 μg) of AT or without the inhibitor for 20 min in 15 mM Tris-HCl, pH 7.4, 5 mM $MgCl_2$, and 5 mM $CaCl_2$ at room temperature. Then 15 μg of pure plasma FN was added (final volume 200 μl) and the reaction mixtures incubated for 8 h at room temperature with end-over-end rotation. The reaction mixtures were centrifuged and 30-μl portions were immunoblotted with FN antiserum. *Lane MW*, molecular weight markers; *lane 1*, undegraded FN; *lane 2*, 8 h digest of FN; *lane 3*, 8 h digest of FN in the presence of 5 μg AT; and *lane 4*, 8 h digest of FN in the presence of 20 μg AT.

increased in the presence of inhibitor and 2) certain lower molecular weight FN fragments between 25 and 45 kDa were not produced in the presence of inhibitor. Antichymotrypsin and α2-antiplasmin also protected FN from degradation (data not shown) but to a lesser extent when compared to AT. Based on these results, it appears that AT was the salient serpin capable of protecting FN from degradation by chronic wound fluid serine proteinases.

Elastase Levels Were Ten- to Fortyfold Higher in Chronic Wound Fluids Compared to Acute Wound Fluids Because elastase degrades FN *in vitro* and has been shown to be the responsible enzyme degrading FN in burn wound fluids [8], elastase activity in acute and chronic wound fluids was quantitated. Elastase levels in acute wound fluids (0.1 to 0.3 mU/mg of protein) were comparable to those in NHS (0.15 to 0.17 mU/mg of protein) **(Table I)**. In contrast, chronic wound fluids associated with degraded FN contains ten- to fortyfold more elastase activity (3 to 8 mU/mg protein) **(Table I)**.

AT-Degrading Enzymes Are Present in Chronic Wound Fluids Associated with Degraded AT Because AT is degraded in chronic wound fluids associated with degraded FN and it was shown to protect FN from degradation by chronic wound fluid enzymes, we wished to determine if AT degrading enzymes are present in wound fluids associated with degraded AT. To address this question, wound fluid proteins were coupled to BZ-6B and incubated with NHS as the source of AT. As shown in **Fig 7**, BZ-6B complexed with proteins from NHS, and AW2 did not degrade AT (**Fig 7**, lanes 2–5 of panels NHS and AW2). In contrast, degradation of AT readily occurred with beads containing proteins from chronic wounds (**Fig 7**, lanes 2–5 of panels CW3, CW6, CW5, and CW7). AT was detected as a 50-kDa protein at 0 min of incubation of the serum with BZ-6B coupled with chronic wound fluids (lane 1). Increasing the incubation time to 30 min (lane 2), 2 h (lane 3), 5 h (lane 4), and overnight (lane 5) resulted in the appearance of 40- and 35-kDa fragments. AT antibodies also

Table I. Elastase Activity in Acute and Chronic Wound Fluids[a]

Sample	Elastase Activity (mU/mg Protein)
NHS	0.19
AW1	0.17
AW2	0.33
AW3	0.30
AW4	0.20
CW2	5.26
CW3	2.75
CW4	3.70
CW5	7.00
CW6	5.15
CW7	4.15
CW8	3.65
CW9	4.75

[a] The assay conditions and the measurement of the enzyme are described in *Materials and Methods*. The values represent the average of two determinations, which differed between 0 and 7 percent.

label a 6-kDa (marked by arrowheads in CW6 and CW7) and an 8-kDa polypeptide in serum proteins after an overnight incubation (lane 5). The degradation of AT by enzymes from chronic wound fluids was further evidenced by a gradual decrease in the molecular size of AT with incubation time (lanes 2–4) and the complete disappearance of the inhibitor in the serum after overnight incubation (lane 5). These data clearly demonstrate that chronic wound fluids with degraded AT also contain enzymes capable of degrading AT.

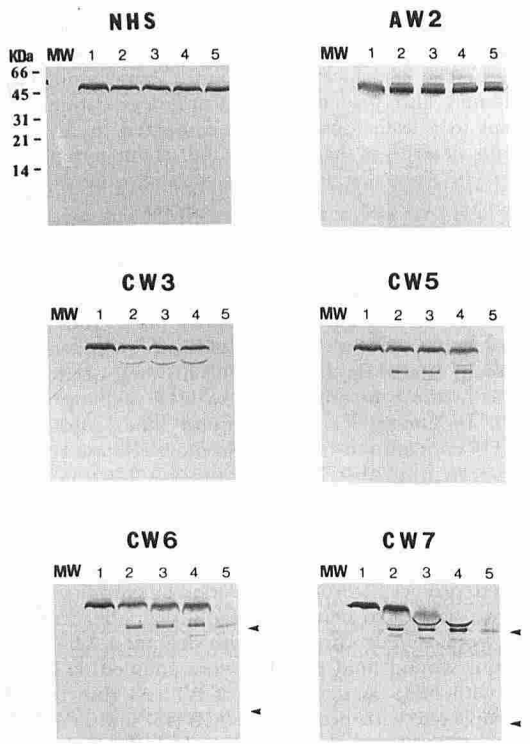


Figure 7. Chronic wound fluids contain enzymes capable of degrading human AT. Proteins from NHS, acute wound (AW) fluid-1, and CW3, CW5, CW6, and CW7 were coupled to BZ-6B and incubated with 400 μg human serum proteins as described in *Materials and Methods*. Supernatants of the reaction mixtures were collected at 0 min (*lane 1*), 30 min (*lane 2*), 2 h (*lane 3*), 5 h (*lane 4*), and 20 h (*lane 5*) and were immunoblotted with polyclonal antibodies to human AT.

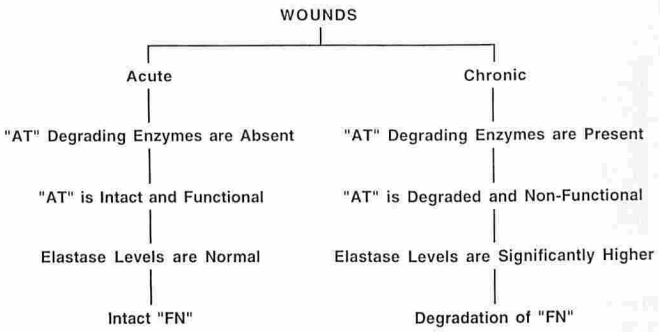


Figure 8. Predicted sequence of events leading to FN degradation in chronic wounds.

DISCUSSION

The results of this series of experiments presents an interesting hypothesis that explains why FN is degraded in chronic wounds **(Fig 8)**. The chain of events begins with the expression of one or more enzymes in chronic wounds associated with degraded FN, but not in acute wounds, which can degrade AT. The degraded AT in chronic wound fluids was non-functional. Elastase, the physiologic target of AT, was 10 to 40 times higher in chronic wound fluids. Thus, the non-functional AT fragments are incapable of inhibiting the serine proteinases (neutrophil elastase) that are responsible for FN degradation in chronic wounds.

We were able to demonstrate the existence of AT- and FN-degrading enzymes in wounds associated with degraded FN and AT but not in wounds associated with intact FN and AT. We have also been able to demonstrate that the degradation of FN by chronic wound fluid enzymes was effectively inhibited by AT. These observations were possible because of the use of BZ-6B to bind wound fluid proteins including enzymes. BZ-6B are covalently linked with a trypsin inhibitor, P-aminobenzamidine, and have been used for purification and/or separation of trypsin, bovine thrombin, urokinase, human enterokinase, native plasminogen, kallikrein, prekallikrein, collagenase, and clostripain [14–16]. Although BZ-6B are covalently associated with a trypsin inhibitor, the enzymes from chronic wound fluids were active and have reproducibly degraded both FN and AT. Coupling the wound fluid proteins to BZ-6B not only eliminated the need to account for the existing FN and AT fragments, but also enabled us to examine the degradation of FN and AT by immunoblotting and silver stain procedures. In addition, any proteinase inhibitors of the wound fluids would not interfere with the ability of the wound fluid enzymes to degrade FN and AT.

Our results suggest that the degradation of AT, by as yet unidentified enzymes, leads to loss of this inhibitor in wounds that eventually feature extensive FN degradation. Degradation of AT was observed in patients with pulmonary emphysema [17–20] and rheumatoid arthritis [21–23]. A number of enzymes of mammalian and bacterial origin degrade AT *in vitro*. Of the mammalian enzymes, MMP-8 [24–26], MMP-1 [27], MMP-3 [28,29], MMP-7 [29], and MMP-9 [30] have been shown to contain AT-degrading activity. Mouse macrophage elastase, a neutral metalloproteinase [31] and human polymorphonuclear leukocyte collagenase [32] also degrade AT. Mast *et al* [33] compared the rates of degradation of AT, antichymotrypsin, and antithrombin III by MMP-1, MMP-2, and MMP-3 and suggested that MMP-3 is a more likely physiologic enzyme-inactivating serum serpins. More recently, Zhang *et al* [30] reported that MMP-7 degraded native as well as oxidized AT four times faster than MMP-3. In addition to MMPs, cathepsin L, a lysosomal thiol-proteinase, also degrades AT [34].

A number of bacterial enzymes have also been reported to degrade AT. These proteinases include metalloproteinase from *Seretia marcescens* [35], serine-, cysteine-, and metalloproteinases

from *Staphylococcus aureus* [36], and a neutral metalloproteinase/elastase from *Pseudomonas aeruginosa* [37]. It has been shown that metabolites derived from *S. aureus* and *P. aeruginosa* impede wound healing [38]. In particular, the presence of *P. aeruginosa* has been found to correlate with non-healing necrotic ulcers [39]. Potempa *et al* [36] proposed that the inactivation of AT by pathogenic bacteria is an effective mechanism for altering AT-neutrophil elastase balance in blood to favor free enzyme and increase plasma protein degradation associated with septicemia. We observed that chronic wound fluids with extensive FN degradation contain a 42-kDa gelatinolytic enzyme (Rao, unpublished data). It will be interesting to investigate if this enzyme is responsible for the degradation of AT in chronic wounds.

Degradation of AT may also contribute to chronic inflammation associated with non-healing chronic wounds. Banda *et al* [40] demonstrated a novel function for the proteolytically modified AT. In contrast to native AT, mouse macrophage elastase-treated AT undergoes proteolysis and acts as a potent chemotactic factor for human neutrophils. Thus, proteolysis of AT can contribute to prolonged inflammation by recruiting more inflammatory cells to the sites of chronic wounds. Inflammatory cells are also the primary source of elastase, which has been shown to degrade FN *in vitro* [8,41]. Although our experiments do not precisely identify elastase as the enzyme responsible for FN degradation, the following findings suggest that it is the responsible enzyme: 1) the enzyme levels were ten- to fortyfold higher in chronic wound fluids compared to acute wound fluids, and 2) it has been identified as the enzyme responsible for FN degradation in burn wound fluids [8]. Collectively, these observations provide us with a basis to propose that the expression of AT-degrading enzymes (neutrophil and/or bacterial) in chronic wounds alters the balance between AT and elastase toward an enzyme-rich environment, resulting in uncontrolled degradation of FN in chronic wounds. A similar imbalance in proteinase-proteinase inhibitor levels leads to increased connective tissue degradation in patients with pulmonary emphysema [18] and rheumatoid arthritis [22].

The observations reported herein may have relevance for the treatment of chronic wounds. Several studies demonstrated a beneficial effect for topical FN on leg ulcers and non-healing corneal ulcers [42–44]. Intact FN may be required for the healing of chronic wounds. The observations reported herein suggest that the degradation of AT precedes the degradation of FN in chronic wounds. We show here that AT effectively prevented the degradation of FN by chronic wound fluid serine proteinases. Therefore, topical AT may inhibit FN degrading enzymes and increase the concentration of intact and functional FN in chronic wounds. In conclusion, we report that chronic wound fluids associated with extensively degraded FN are associated with degraded AT and AT-degrading enzymes. AT inhibited the degradation of FN by chronic wound fluid enzymes, suggesting that this inhibitor regulates FN degradation in chronic wounds.

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